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Integrating Machine Learning Clustering into PVT Based Reservoir Fluid Characterization Workflows

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Abstract

The characterization of reservoir fluids is pivotal to multiple facets in the life cycle of a hydrocarbon bearing asset. The process of reservoir fluid characterization starts with the collection of representative samples which are then subjected to laboratory testing for the measurement of Pressure-Volume-Temperature (PVT) properties as well as other properties such as the fluid and isotopic composition. Subsequently, an equation of state may be developed from the measured data. Ultimately, the equation of state provides the capability to predict other fluid properties that may not have been physically measured in the laboratory. The process of sampling and conducting laboratory analysis are time consuming and have an associated cost. This paper presents a novel process by which the properties of a characterized ‘parent’ sample may be utilized to infer the properties of a new ‘child’ sample based on compositional data and saturation pressure measurements from the ‘child’ sample.

Introduction

Understanding the properties of reservoir fluids is vital for optimizing the performance of hydrocarbon assets at every stage, from exploration to development. Some examples highlighting the importance of reservoir fluid characterization include:

- Fluid characterization plays a critical role in estimating hydrocarbon reserves, and its accuracy has a ripple effect on reserve bookings, financial reporting, and investment decisions, making it a key factor in a company's overall success.
- Comprehensive understanding of in-situ fluid properties enables the development of tailored well placement, drilling and completion strategies that optimize resource recovery.
- Accurate and timely assessment of fluid composition allows operators to develop effective plans for treating, processing, and transporting produced fluids, thereby reducing safety risks, environmental impacts, and operational costs.

The current approach to reservoir fluid characterization starts with the acquisition of accurate and representative fluid samples from the reservoir via sampling operations. These samples undergo laboratory studies, where essential fluid characteristics related to PVT relationships are measured under simulated

reservoir conditions. The measured data are then used to develop an equation of state (EOS) model that describes the fluid's thermodynamic behavior. With this model, unmeasured properties can be computed across varying pressure and temperature ranges. Using these computed fluid properties, the reservoir fluid can be categorized as either oil or gas. Additionally, critical characteristics such as API gravity, Gas-Oil Ratio, and formation volume factor can be determined at conditions of interest. Finally, the characterized fluid properties are integrated into numerical models to forecast reservoir performance, informing strategic decisions about well placement, production optimization, and effective reservoir management practices.

Throughout this process, repeated analyses and simulations allow for continuous refinement of our understanding of reservoir fluid behavior. Moreover, ongoing technological advancements and deeper insights into fluid dynamics continually enhance this established methodology for thermodynamic based reservoir fluid characterization. As mentioned earlier, this process is time consuming and has an associated cost which becomes even more significant when considering dealing with sampling in harsh environments such as ultra-sour reservoirs and deep-water developments.

This paper presents a novel process by which the properties of a characterized 'parent' sample may be utilized to infer the properties of a new 'child' sample based on only compositional data measurements and the saturation conditions of the 'child' sample. An un-supervised learning model, like the k-means algorithm, is applied to a group of 'parent' data sets for the purpose of segregating the datasets into clusters of related samples. The 'parent' data set should be as detailed and accurate as possible. By detailed, it is meant that there should be multiple data points from a particular reservoir, multiple data points from a field, and multiple data points from a region. By accurate, it is meant that a thorough QA/QC process is in place to identify and exclude unrepresentative samples. These data should include results from routine laboratory testing such as compositional analysis, constant composition experiments (CCE) and constant volume depletion tests (CVD). Data from isotope analysis has been critical in supporting or discounting the results of this process. The study utilizes data from Carbon isotopic compositions of C1-C5 to validate the results of the process.

Part of the uniqueness of this approach are the features utilized for the clustering, which include the following features:

- Concentration of N_2
- Concentration of CO_2
- Concentration of H_2S
- Dryness defined as $C1/(C1+C2+C3+C4+C5)$
- Wetness defined as $[(C2+C3+C4+C5)/(C1+C2+C3+C4+C5)]$
- Balance ratio defined as $[(C1+C2)/(C3+C4+C5)]$
- Character ratio defined as $(C4+C5)/C3$
- Concentration of C7+
- Density of C7+
- Molecular weight of C7+

By applying a clustering algorithm to these features, the various 'parent' samples can be clustered into different classes and should reflect any existing relationships. Thus, if the data from a 'child' sample is introduced into the clustered arrangement, then the 'child' sample will fall into a cluster of 'parent' samples that is most closely related to it. With the establishment of a relationship between a 'child' and 'parent'

sample, the properties of the ‘parent’ sample in conjunction with the saturation conditions of the ‘child’ sample may be utilized in predicting the properties of the ‘child’ sample.

The ethos of this novel process is maximizing the utility of collected data while minimizing the costs associated with sampling and laboratory testing required to measure the ‘child’ sample's properties. This process aims to leverage existing data in predicting the phase behavior of fluids that cannot be tested for reasons such as operational constraints. The process also has application in QA/QC of laboratory test data as well as missing data management in PVT data handling.

Some of the possible applications of this novel process include:

1. **Fluid Property Prediction:** By predicting the characteristics of new ‘child’ samples, such as gas composition, density, and viscosity, this approach provides critical information for optimizing gas processing, transportation, and storage systems, and ensures the safe and efficient handling of hydrocarbons.
2. **Fluid Sampling Optimization:** This approach optimizes fluid sampling programs by selecting the most representative samples and predicting the properties of new ones, ultimately reducing the number of samples needed and enhancing the accuracy of reservoir modeling.
3. **Reservoir Compartmentalization:** This approach enables the detection of reservoir compartmentalization by recognizing patterns and relationships between fluid samples from diverse locations, thereby optimizing recovery outcomes.
4. **Gas Reservoir Characterization:** This approach enables the clustering of fluid samples from various wells and locations, identifying patterns and relationships that inform predictions for new samples, and ultimately enhancing reservoir modeling, gas-in-place estimates, and field development strategies to optimize recovery and production.
5. **Gas Reservoir Modeling:** This method strengthens the accuracy of gas reservoir models by integrating the predicted behavior of new ‘child’ samples and the connections with ‘parent’ fluid samples, resulting in more reliable predictions, better decision-making, and improved reservoir management.

Methods, Procedures, Process

Data and machine learning analysis

"No data is clean, but most is useful.", a quote by Dean Abbott, Co-founder and Chief Data Scientist at SmarterHQ is an apt saying that highlights the primacy of data in all analytical undertakings. As such, our approach starts with the data and with exploratory data analysis (EDA) process to be more precise. EDA is a structured methodology that enables data scientists to investigate, examine, and visualize datasets, uncovering their key features, trends, and correlations, and laying the groundwork for further analysis and modeling¹.

In this case, EDA relied heavily on dimensionality reduction to simplify complex high-dimensional data. The benefits of dimensional reduction were captured in this approach by incorporating the concept of Chromatographic gas ratios presented by Harworth et al. (1984) and Harworth et al. (1985). These ratios include the following;

- Dryness defined as $C1/(C1+C2+C3+C4+C5)] \times 100$
- Wetness defined as $[(C2+C3+C4+C5)/(C1+C2+C3+C4+C5)] \times 100$
- Balance ratio defined as $[(C1+C2)/ (C3+C4+C5)]$
- Character ratio defined as $(C4+C5)/C3$

By applying dimensionality reduction techniques, such as using mud logging ratios like Dryness and Wetness, the number of features utilized in the analysis could be minimized abating the occurrence of overfitting while making the dataset easier to visualize and understand. This process enabled the identification of connections and relationships that were previously hidden by the excessive variability of the data. The dryness is indicative of gas origins, migration and accumulation. Marked changes in the dryness provide a useful clue in understanding the origin of potential secondary changes in gas reservoirs². The wetness ratio shows the proportion of heavy components in a well stream while the balance ratio is inversely proportional to the wetness ratio and confirms gas production potential³. The character ratio is presented as a quantity which provides confirmation of reservoir fluid character in certain ambiguous situations. EDA also included a simple 1-D plot to establish preliminary relationships and identify potential ‘parent’ clusters. This plot also assisted in performing a QA/QC of the available data.

An isolation forest algorithm was utilized in assessing the data for outliers and removing those outliers identified as unrepresentative to ensure that the clustering process was not in any way distorted leading to the occurrence of misclassification. The variables were normalized by standardization to manage the effect of scale invariance as well as reduce the impact of any present outliers. All the variables were also assessed for normality with QQ-plots. A Box-Cox transformation was utilized to transform the non-normally distributed data to fit a normal distribution as much as possible. The variables were then assessed for collinearity using a correlation plot to minimize the number of variables used and minimize the level of noise associated with the analysis.

The concept of clustering algorithms has a rich history that spans nearly a century, with its roots tracing back to the early 20th century in the fields of anthropology and psychology. In the 1930s, clustering emerged as a method to simplify complex typologies of cultures and individuals in anthropology, with notable contributions from Driver and Kroeber in 1932. Around the same time, psychologist Cattell began using clustering in 1943 to classify traits in personality psychology, aiming to organize and analyze meaningful information from psychological data⁴.

Overall, the development of clustering algorithms has been marked by improvements in identifying similarity, handling larger datasets, and adapting to complex data types. The field continues to evolve with the introduction of new techniques and algorithms, particularly in the area of deep learning.

One popular clustering algorithm is k-means, which groups similar data points together based on their attributes. To illustrate, imagine sorting colored balls into separate baskets, where each basket represents a cluster. The process involves randomly placing the balls into ‘k’ baskets, then moving each ball to the closest matching basket based on its color. This process is repeated until no more balls need to be moved. K-means is advantageous due to its ease of implementation, scalability, and speed. It's particularly effective for discovering spherical clusters with distinct centroids. However, it has some limitations, including the need to specify the number of clusters (‘k’) beforehand, which can be challenging. Additionally, k-means struggles with non-linearly separable data, outliers, and noise, leading to suboptimal results. The initial placement of centroids also affects the final outcome, requiring careful initialization.

To test this novel process, 120 gas samples from 6 fields and 8 reservoirs were captured with substantial variations in spatial and vertical sampling points to comprehensively capture the existing heterogeneities, with 20% of the samples being subjected to routine laboratory testing. Routine laboratory testing covers compositional analysis, CCE, CVD, and separator tests. Data pre-processing includes a thorough QA/QC assessment to ensure the utilized data were competent and representative. The k-means clustering algorithm was trained with only the ‘parent’ data. The average distortion and silhouette values were utilized in identifying the appropriate model hyperparameters. With the identified hyperparameters, the ‘child’ sample data was then included in the total dataset to identify which ‘parent’ cluster the ‘child’ sample of interest would fall into. The cluster relationship between ‘parent’ and ‘child’ was then assessed by measuring the ability to utilize ‘parent’ cluster properties to predict ‘child’ sample properties. The properties of interest include the GOR, relative volume, density, z-factor amongst others. The performance of the ‘parent’ cluster

model in predicting the characteristics of the ‘child’ sample was measured by quantifying the Mean Absolute Percentage Error (MAPE) between the measured property value of the ‘child’ sample and the predicted property value.

Carbon Isotope Analysis

Pressurized samples of natural gas were collected from different reservoirs through production tests, drill stem tests or downhole samplers and analyzed for chemical composition. The collected gas samples were analyzed using a gas chromatograph with multiple detectors to measure concentrations of both hydrocarbon and non-hydrocarbon gas components. Data reproducibility for the measured samples was better than 0.1%. Carbon isotope analysis for the gas samples was conducted using GC equipped with a Finnigan GCC III combustion furnace interface coupled to a stable isotope ratio mass spectrometer. Carbon isotope ratios were measured for individual hydrocarbon compounds from methane to pentane, and for CO₂. Measurements are reported in delta notation as part-per-thousand (‰) variations in atomic ¹³C/¹²C relative to the Vienna Pee Dee Belemnite, VPDB international reporting standard:

$$\delta^{13}\text{C}(\text{‰}) = 1000 * (R_{\text{SAMPLE}} / R_{\text{VPDB}} - 1), \text{ where } R = {}^{13}\text{C} / {}^{12}\text{C}.$$

Negative $\delta^{13}\text{C}$ values indicate enrichment of ¹²C relative to the VPDB standard, whereas positive values indicate enrichment of ¹³C. Analytical precision, determined by replicate analysis of samples, was generally better than ± 0.2 . Repeated analysis of the in-house gas standard yielded a reproducibility better than 0.3 ‰.

Results, Observations, Conclusions

The QA/QC process was utilized in identifying and removing non-representative samples. [Figure-1](#) is an example of some of the QA/QC tools employed. The results of testing performed using 2 ‘child’ samples from Field-X (Reservoir-5) and from Field-Y (Reservoir-4) are now presented. These tests required the characterization of both ‘child’ samples based on just the compositional data and saturation conditions. Please note that CCE, CVD, and separator test data for each ‘child’ sample were measured but sequestered from the analysis process and utilized for the sole purpose of validation/comparison. The results of the identified relationships between ‘parent’ and ‘child’ samples were compared with data from isotope tests for confirmation.

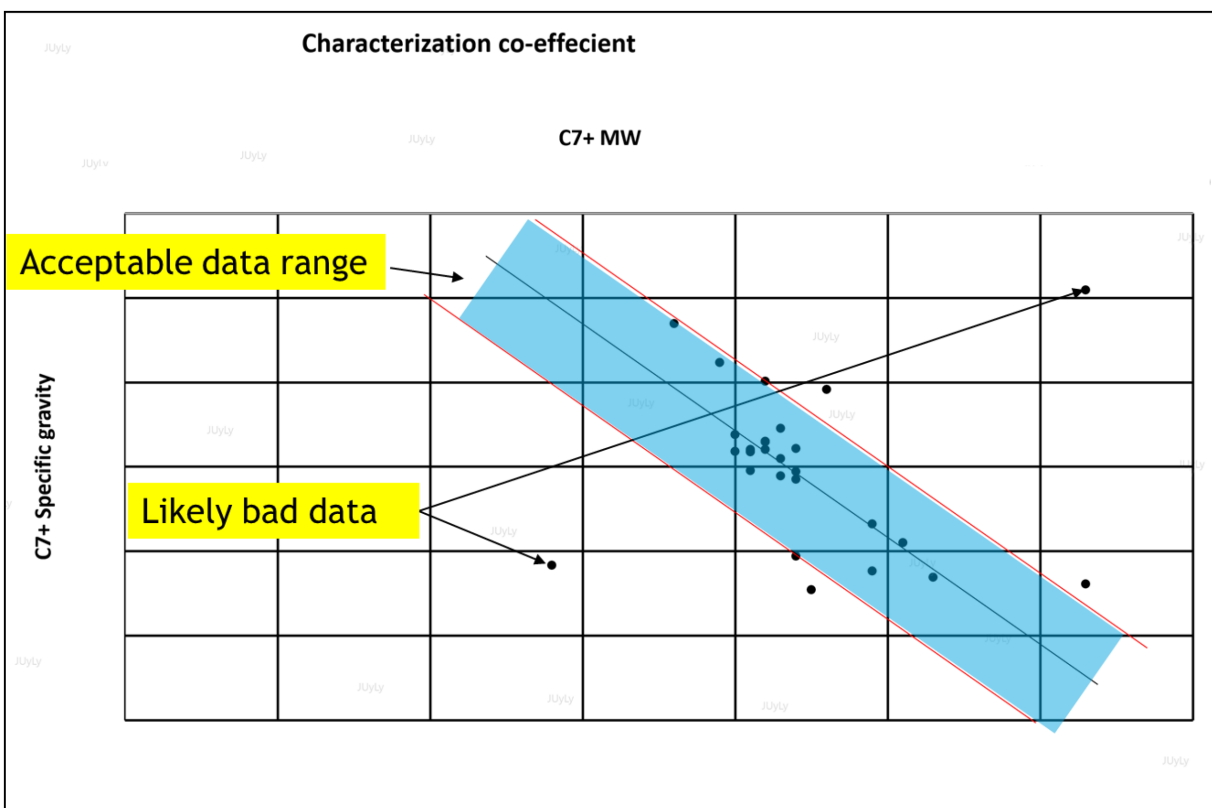


Figure 1—Plot of sample molecular weight vs. C7+ specific gravity as a QA/QC tool

The EDA showed that some of the features did exhibit outliers and did not display a normal distribution. For instance, [Figure-2](#) shows the dryness with several outliers and a left skewed distribution. After excluding the outliers assessed to be unrepresentative and applying a Box-Cox transform, the figure shows the distribution to be closer to normality.

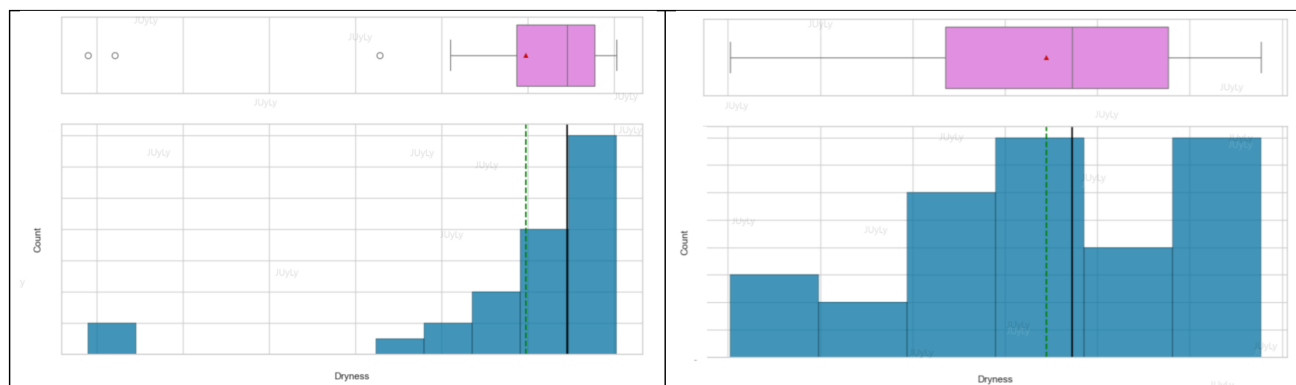


Figure 2—EDA results pre and post data transformation

[Figure-3](#) shows the correlation between all features included in the analysis. What is readily clear is that the Density of C7+ and dryness are approximately colinear and the Molecular weight of C7+ and balance ratio are also approximately colinear. As such, the variables Density of C7+ and Molecular weight of C7+ were excluded from the analysis to prevent overfitting.

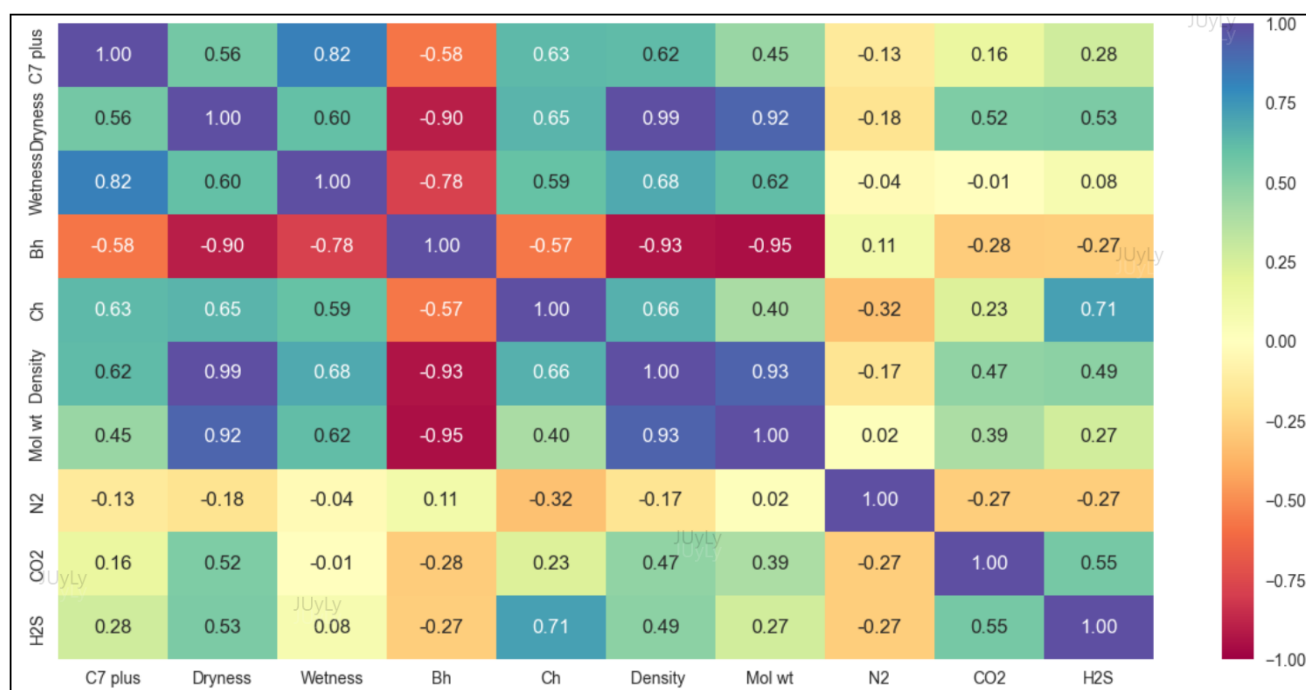


Figure 3—Plot of feature variables to identify correlation

Figure-4 is a 1-D plot of dryness vs depth. This plot shows that Field-1 has 3 reservoirs (Reservoirs-1, 2, & 6). Field-2 has 4 reservoirs (Reservoir-1, 5, 7, & 8). Field-3 has 1 reservoir (Reservoir-3). Field-4 has 4 reservoirs (Reservoir-2, 6, 7, & 8). Field-6 has 5 reservoirs (reservoir-1, 2, 3, 7, & 8). A reservoir number showing up in different fields indicates a common reservoir identified in those fields. This plot shows that samples from Field-2 (reservoir-5 & 7) show dissimilar characteristics for the respective reservoirs. Samples from both reservoir-7 and reservoir-5 exhibit properties that differentiate the respective reservoir samples into 2 groups. This is an indication of possible changes in the origin, accumulation, migration or potential secondary changes to the reservoir fluid that may have occurred and affected half the respective reservoir samples while leaving the others unaffected. This indicates possible discontinuities in both reservoirs as well as the possibility of differing geological processes in each compartment. The 1-D plot shows that the samples of interest from Field-X (Reservoir-5) and Field-Y (Reservoir-4) do not fall in any groups identified.

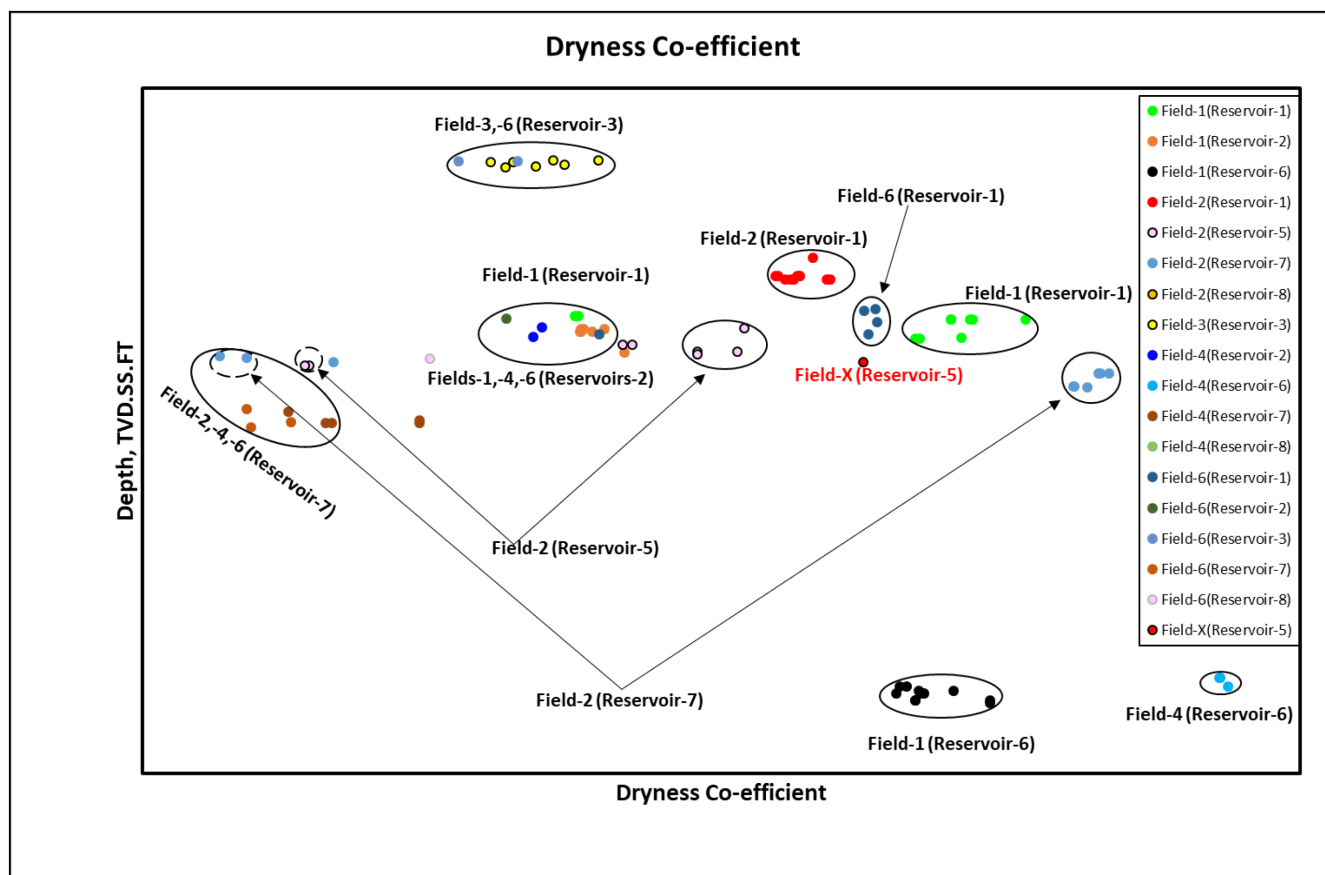


Figure 4—1-D plot of Dryness vs. Depth as an initial assessment tool

After sanitizing the data, a k-means algorithm was used to cluster the ‘parent’ sample data from every field and reservoir. Using the average distortion and the Silhouette score, the best hyperparameters were identified including the number of clusters as shown in Figure-5 & 6. Data from Field-X (Reservoir-5) and Field-Y (Reservoir-4), the ‘child’ samples, were included in the analysis dataset along with the ‘parent’ sample dataset without making any changes to the original hyperparameters. Figure-7 shows that the ‘child’ sample from Field-X (Reservoir-5) falls in cluster-5 which is related to Field-2 (Reservoir-5), while the ‘child’ sample from Field-Y (Reservoir-4) falls in cluster-2 which is related to Field-4 (Reservoir-4). This figure also shows the ‘parent’ samples that define each of the 6 clusters identified.

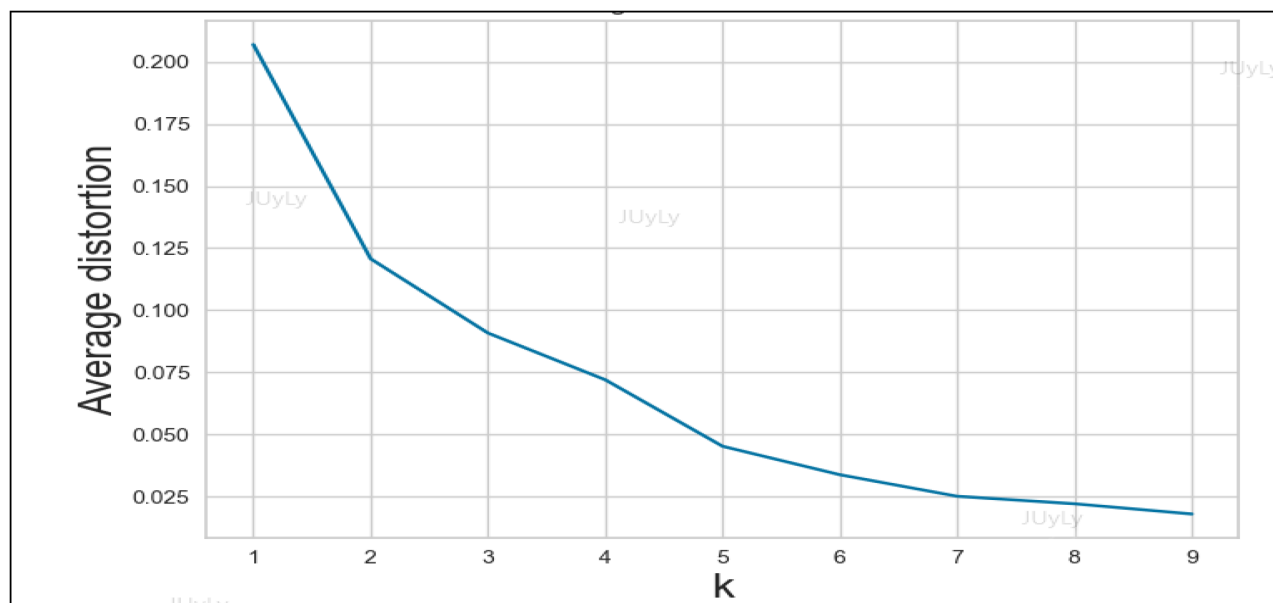


Figure 5—Plot of average distortion vs. cluster count



Figure 6—Plot of Silhouette score

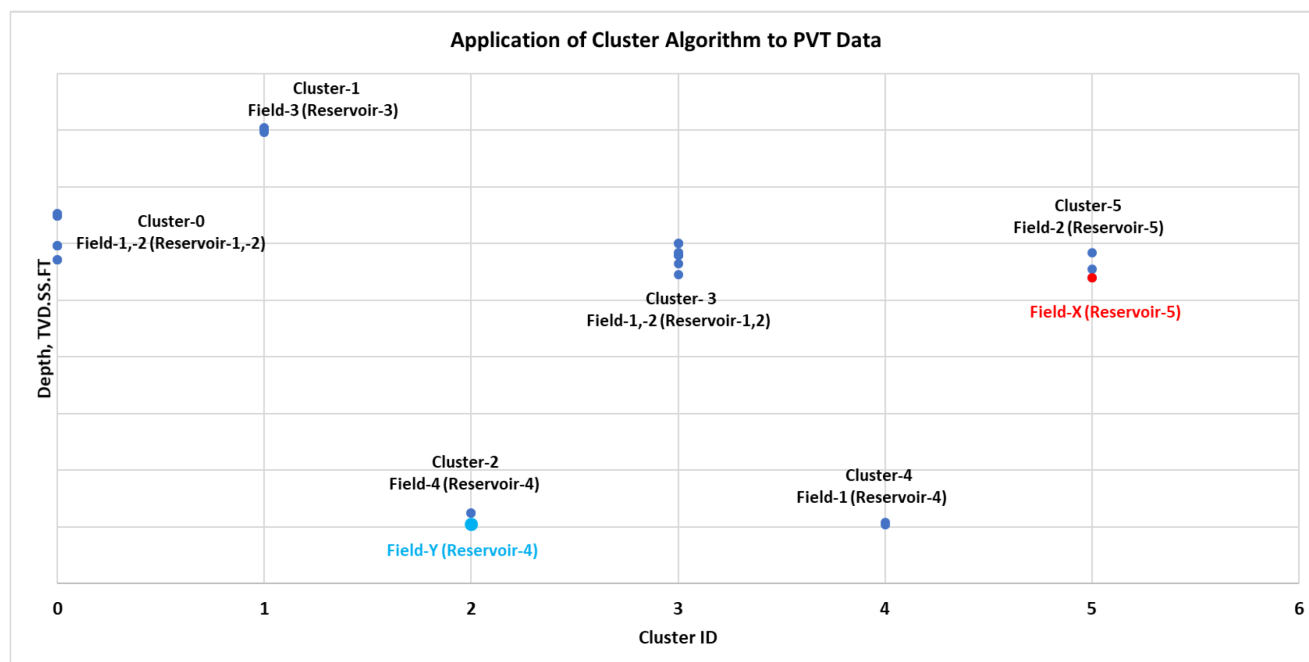


Figure 7—Plot of cluster distribution

With a relationship identified between both ‘child’ samples and the ‘parent’ clusters, the routine laboratory testing data from the ‘parent’ clusters could hypothetically be used to characterize both ‘child’ samples. Using the routine laboratory testing data from Field-2 (Reservoir-5), an equation of state (EOS) was developed. The compositional data of the ‘child’ sample, from Field-X (Reservoir-5), along with its saturation pressure at the reservoir temperature were transposed onto the EOS. No other changes were made. Figure-8 shows the match between the predicted properties of the sample from Field-X (Reservoir-5) compared against the measured property as a means of validation. The error associated with the novel process method is shown on Table-1. Figure-9 shows a comparison between the measured compositional results of the CVD test against the predicted values for different pressure steps. The highest error was observed in the predicted values of 2-phase Z factors with a MAPE of 9.43%. This was mainly due to about 86% of the errors coming from the last 2 pressure steps of the experiment.

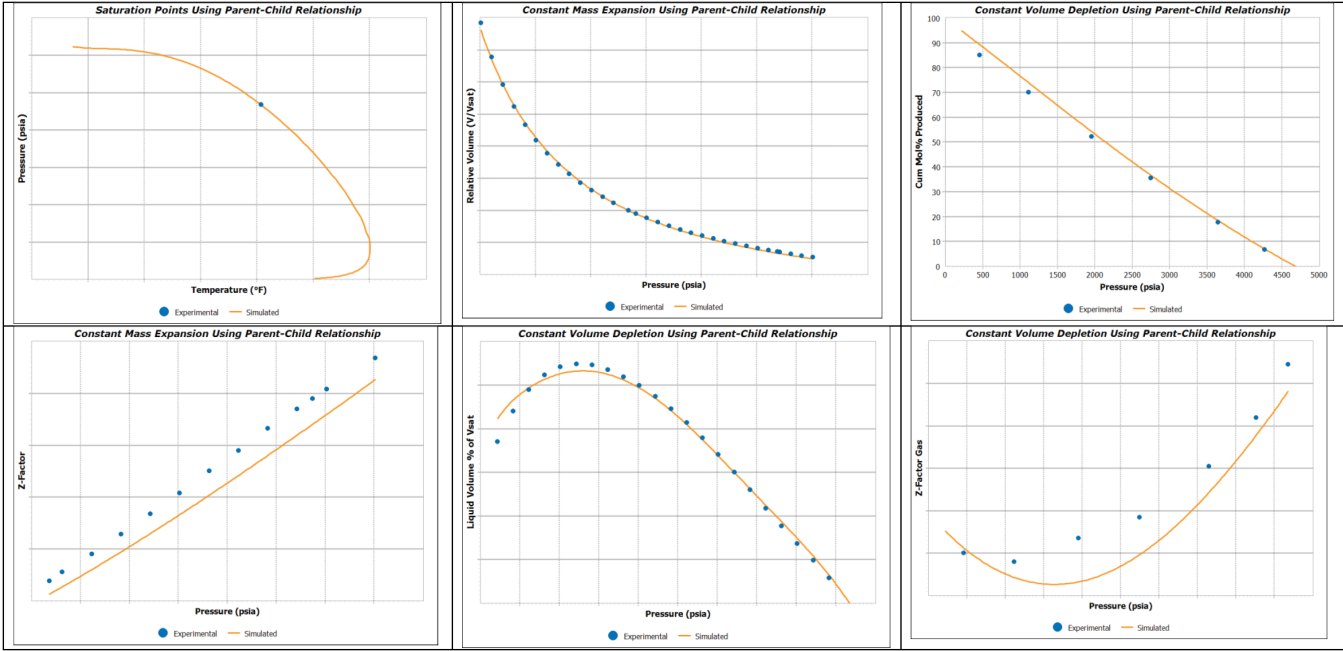


Figure 8—Plot of measured and predicted properties for ‘child’ sample from Field-X (Reservoir-5)

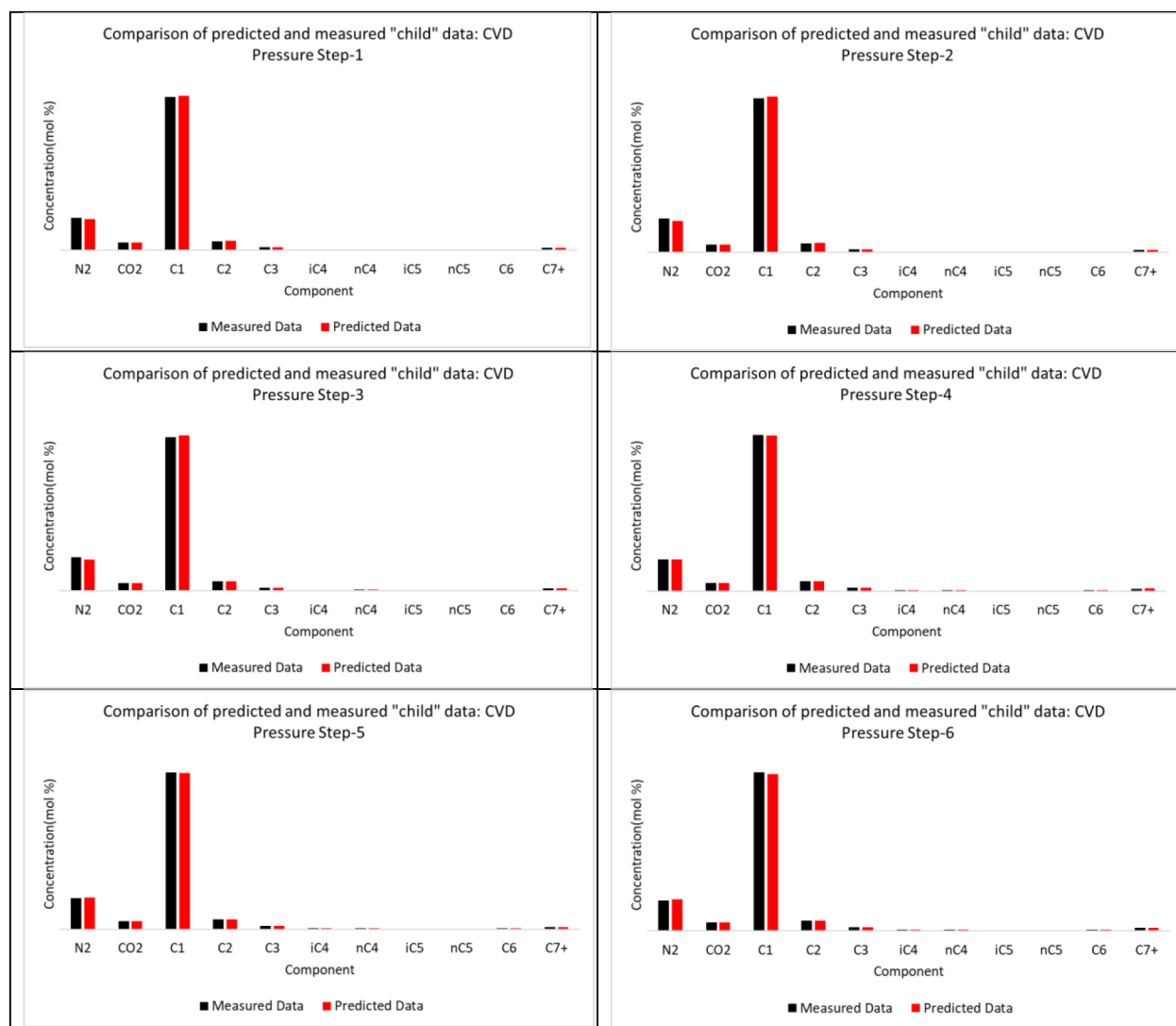


Figure 9—Plot of measured and predicted CVD data for ‘child’ sample from Field-X (Reservoir-5)

Table 1—Prediction error for ‘child’ sample from Field-X(Reservoir-5)

Experiment	MAPE(%)
Relative Volume	0.77
Z-Factor-CME	1.73
Liquid Volume % of Vsat	2.94
Cumulative Mol% Produced	4.49
Z-Factor Gas	1.14
Z-Factor Two Phase	9.43
Produced Well Stream composition(Pressure step-1)	3.04
Produced Well Stream composition(Pressure step-2)	6.98
Produced Well Stream composition(Pressure step-3)	6.71
Produced Well Stream composition(Pressure step-4)	5.60
Produced Well Stream composition(Pressure step-5)	4.77
Produced Well Stream composition(Pressure step-6)	2.20

The process was repeated with the ‘child’ sample from Field-Y (Reservoir-4). Having identified this sample as being related to ‘parent’ samples from cluster-2. Routine laboratory testing data from cluster-2 was utilized to develop an EOS using the compositional data from the ‘child’ sample along with its saturation pressure. **Figure-10** shows the match between the measured data and the predicted properties of the ‘child’ sample from Field-Y (Reservoir-4) using an EOS developed from the ‘parent’ sample data. **Table-2** shows the highest MAPE of 15% was observed in the predicted composition measured at pressure step-6. **Figure-11** shows how the predicted and measured compositions align.

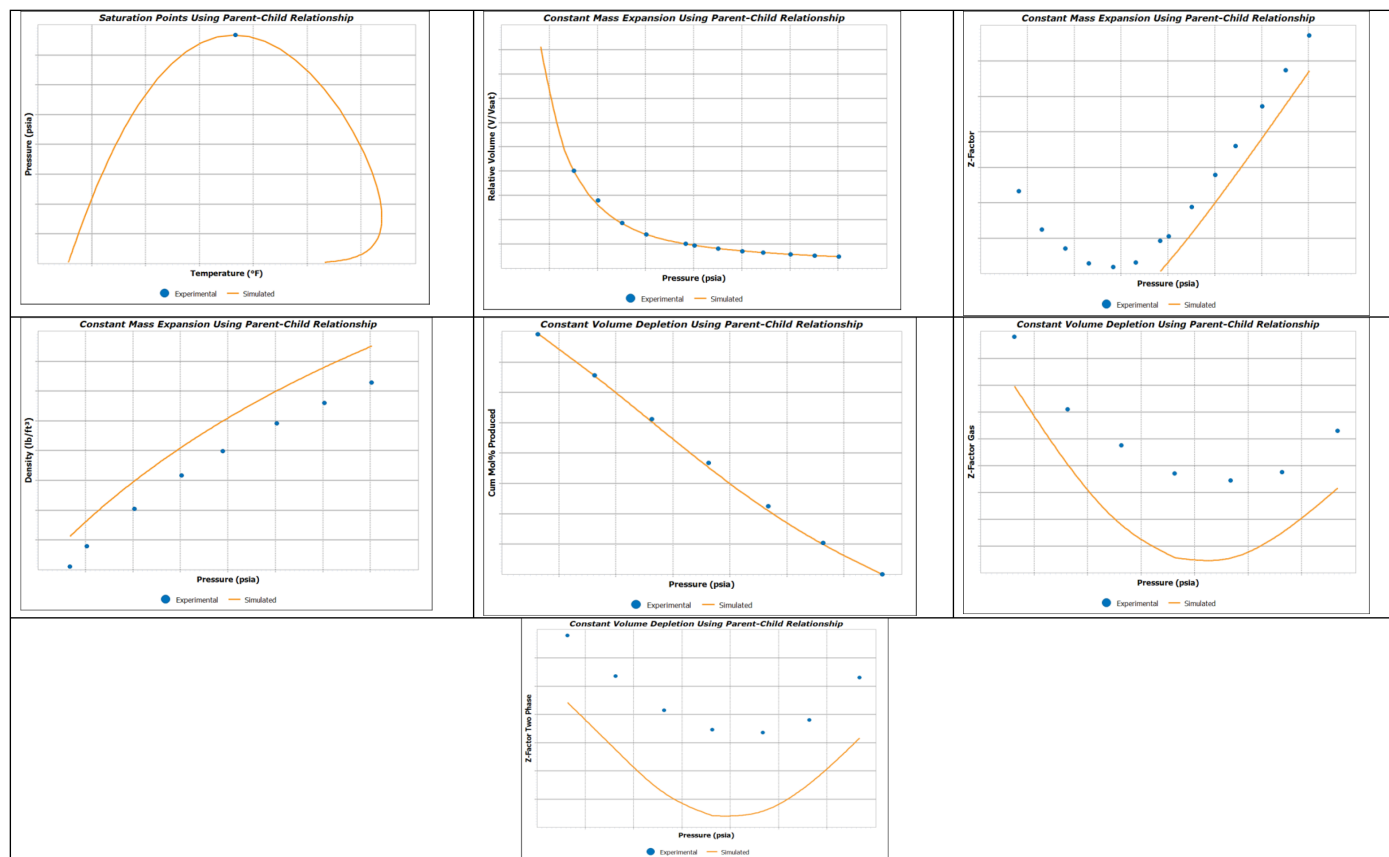


Figure 10—Plot of measured and predicted properties for ‘child’ sample from Field-Y (Reservoir-6)

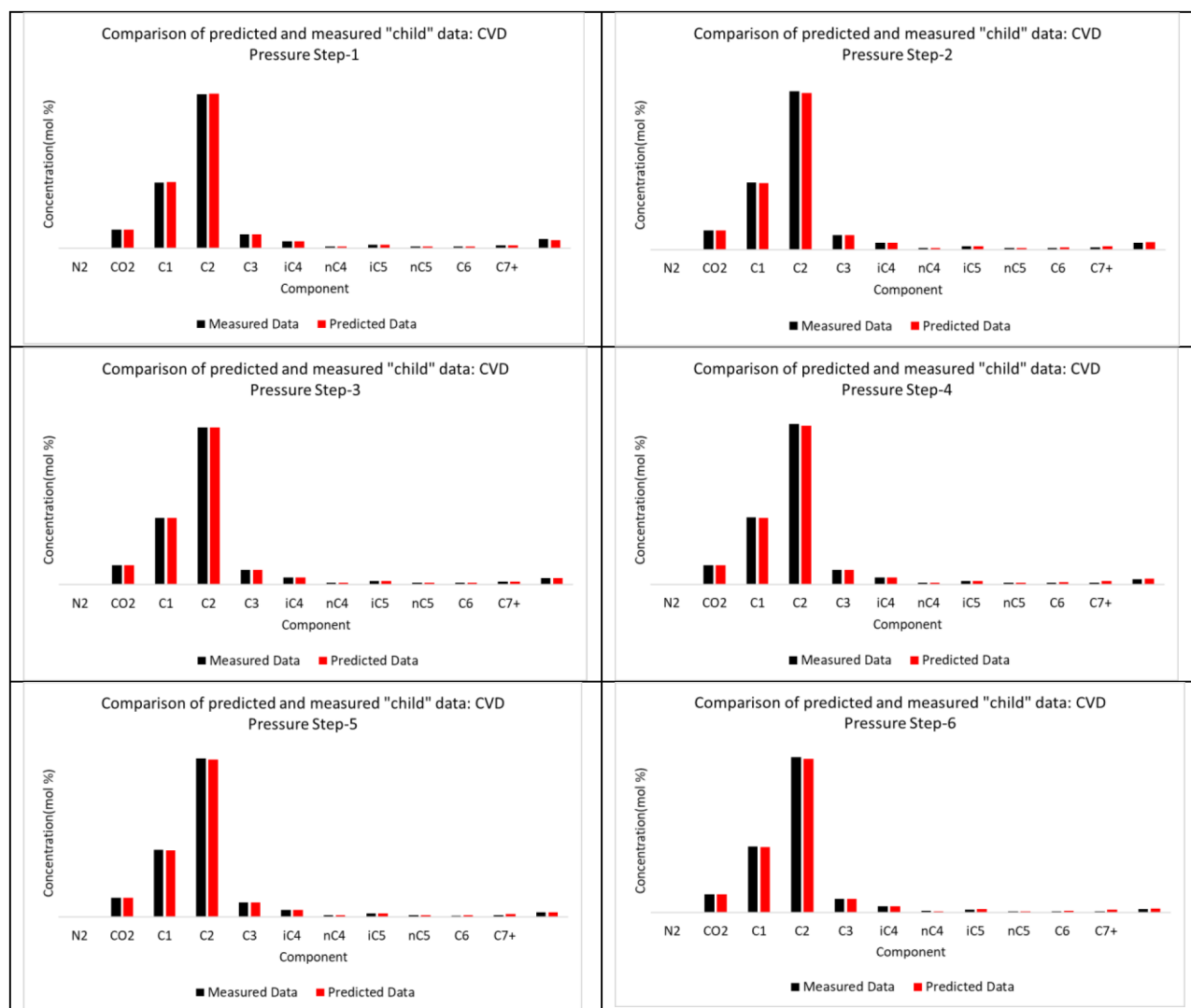


Figure 11—Plot of measured and predicted CVD data for 'child' sample from Field-Y (Reservoir-6)

Table 2—Prediction error for 'child' sample from Field-X(Reservoir-5)

Experiment	MAPE(%)
Relative Volume	0.98
Z-Factor-CME	4.21
Density	5.01
Cumulative Mol% Produced	2.69
Z-Factor Gas	5.90
Z-Factor Two Phase	6.24
Produced Well Stream composition(Pressure step-1)	1.31
Produced Well Stream composition(Pressure step-2)	7.97
Produced Well Stream composition(Pressure step-3)	12.77
Produced Well Stream composition(Pressure step-4)	13.46
Produced Well Stream composition(Pressure step-5)	12.75
Produced Well Stream composition(Pressure step-6)	15.00

Figure-12 shows Carbon isotope type curves constructed for compounds from methane through *n*-pentane can be used to group the gas samples into distinct families, and correlate them to their source rocks^{5,6}. Thus, gas samples generated from different sources often have distinctive carbon isotope type curves. In this study, carbon isotopic analysis was used to determine the different gas families (i.e., parents) in order to validate the presented methodology. This plot identified at least six different gas families ('parent') with significant differences in the carbon isotopic signatures. This is in line with the clustering derived from the k-means algorithm.

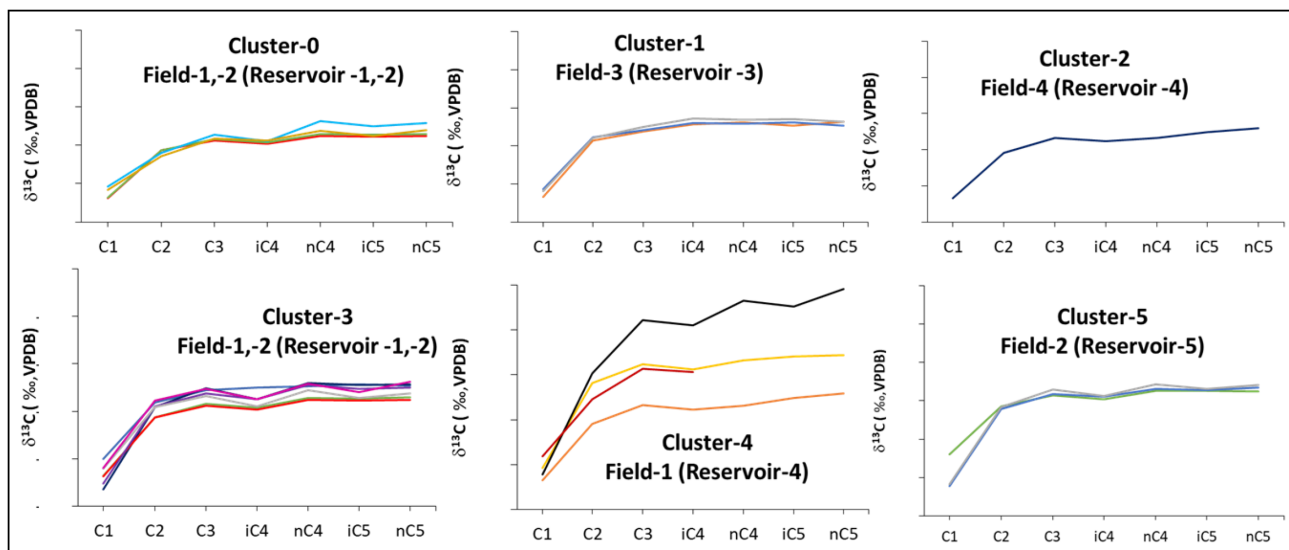


Figure 12—Compound specific carbon isotopic composition (C1-C5) of the studied gases used to genetically classify different gases into multiple clusters.

A significant observation made during these tests is the impact of data quality on the quality of prediction results. The 'parent' sample data quality has to be assessed thoroughly as it impacts the prediction of the 'child' sample properties.

In conclusion, this paper presents a novel approach to reservoir fluid characterization that leverages the relationships between 'parent' and 'child' fluid samples to predict the properties of new, uncharacterized 'child' samples. By applying a k-means clustering algorithm to a comprehensive dataset of fluid samples from multiple fields and reservoirs, we demonstrate the ability to identify patterns and relationships that enable accurate predictions of fluid properties, such as gas composition, density, and viscosity. The results of our testing show that this approach can be utilized to optimize fluid sampling programs, detect reservoir compartmentalization, and improve the accuracy of gas reservoir models. The findings of this study highlight the criticality of data quality and the need for thorough assessment of 'parent' sample data to ensure accurate predictions. Overall, this novel approach has the potential to revolutionize the field of reservoir fluid characterization, enabling more accurate and efficient decision-making in the oil and gas industry. By maximizing the utility of collected data and minimizing the costs associated with laboratory testing, this method can help to optimize recovery outcomes, reduce costs, and improve the overall efficiency of hydrocarbon production. The next phase of this study includes additional testing to understand how inaccuracies in 'parent' dataset impact prediction of 'child' sample properties and how to minimize these errors.

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